CS2103 / Lab – Assignment 09

Topic: Naïve Bayes Classification

Instructions: Complete all five questions. Implement the **Naïve Bayes classifier from scratch using Python/NumPy**. Built-in Naïve Bayes classifiers are not allowed except for verification.

Dataset: `assignment09_flu_symptoms.csv`(Available on SharePoint)

Q1. Dataset Analysis and Probability Estimation

Load the Flu Symptoms dataset and automatically compute the class priors

$$P(Flu = Y), \qquad P(Flu = N),$$

and all conditional probabilities $P(a_i \mid v_j)$ for each symptom value and class. Display the learned probabilities in a clear table.

Q2. Naïve Bayes Classifier from Scratch

Implement Naïve Bayes classification using

$$Score(v_j) = P(v_j) \prod_{i=1}^{n} P(a_i \mid v_j).$$

For the patient Chills=Y, RunnyNose=N, Headache=Mild, Fever=Y, compute the score for both classes and predict whether the patient has flu. Print all intermediate probabilities used in the prediction.

Q3. Classify Multiple Unseen Patients

Create at least **five new symptom combinations** not directly copied from the dataset.

Use your Naïve Bayes implementation to predict the class of each patient. For every prediction, print

$$Score(Flu = Y), \qquad Score(Flu = N)$$

and the final predicted class.

Q4. Conditional-Independence

Choose any two symptoms A and B and a particular value for each. For both classes $C \in \{Flu = Y, Flu = N\}$, compute

$$P(A, B \mid C)$$

and

$$P(A \mid C)P(B \mid C).$$

Repeat the experiment for at least **three different symptom pairs** and check whether the **empirical data exactly satisfy** the conditional-independence assumption for the selected cases.

Briefly explain the observed differences.

Q5. Probability Sensitivity Challenge

Using your Naïve Bayes implementation, select at least three different symptom combinations and compute

$$Score(Flu = Y) \qquad \text{and} \qquad Score(Flu = N).$$

For each case, change **one symptom value at a time** while keeping all other symptoms fixed.

Observe how the two class scores and final prediction change. Identify which symptom changes have the strongest effect on classification and explain the result using the learned conditional probabilities $P(a_i \mid v_j)$.

Note: Students must be able to explain class priors, conditional probabilities, the Naïve Bayes assumption, probability estimation, classification scores, conditional independence, and observed prediction behavior during the evaluation/viva.